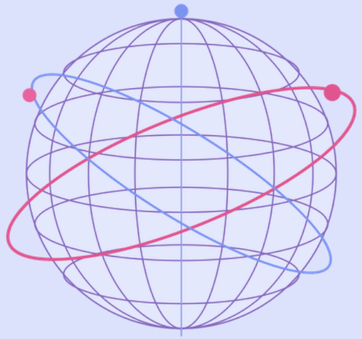
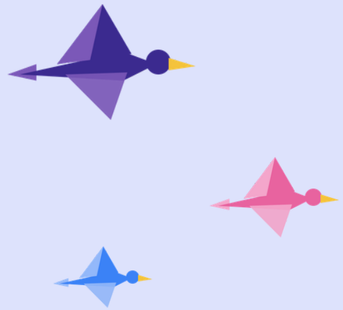
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
QUANTA — Quantum-Assisted Timetabling & Allocation

TRACK
Quantum Cloud based Application

TEAM NAME
Team VTU-QFF-HC-0076

TEAM LEADER
Yash koparde

EMAIL / CONTACT
yashkoparde2022@gmail.com

TEAM MEMBERS
Yash koparde · Melwin Fernandes · Aditya Patange

INSTITUTION / ORGANISATION
Visvesvarayya technological University Belagavi

COUNTRY
India

Problem Statement

THE REAL-WORLD PROBLEM

Imagine you are the chief examination coordinator at a university with 4,200 students, 180 courses, 22 examination halls and 15 time slots across one exam week.

Hard constraints — zero tolerance

No clashes for any student
Seats ≥ students in every exam
No hall double-booking
Every course has a slot and hall

Soft constraints — wellbeing

Back-to-back exams: fatigue
3 exams in a day: overload
Uneven spread across the week

Why existing methods fail

Classical alone: tangled electives blow up runtime
Quantum alone: 100–156 qubits; campus needs tens of thousands

Objectives & Novelty

Objectives

1. **Zero hard violations**
   Clash-free, capacity-safe, fully covered timetable
2. **Cut student fatigue**
   Lower back-to-back and 3+ exams-a-day penalties
3. **Fit NISQ limits**
   Quantum only on the worst ≤ 12-course bottleneck
4. **Prove it honestly**
   Benchmark against CP-SAT, brute force, annealing

Novelty

1. **Quantum proposes, classical disposes**
   Deterministic acceptance gate has the last word
2. **1 qubit per course**
   Perturbation encoding: stay in slot A or move to slot B
3. **Independent auditor**
   Pure-Python scorer — no OR-Tools or Qiskit imports
4. **One pipeline, three backends**
   Ideal sim, noisy Aer, IBM Kingston QPU

Solution & Quantum Advantage

The Hybrid Solution (How It Works)

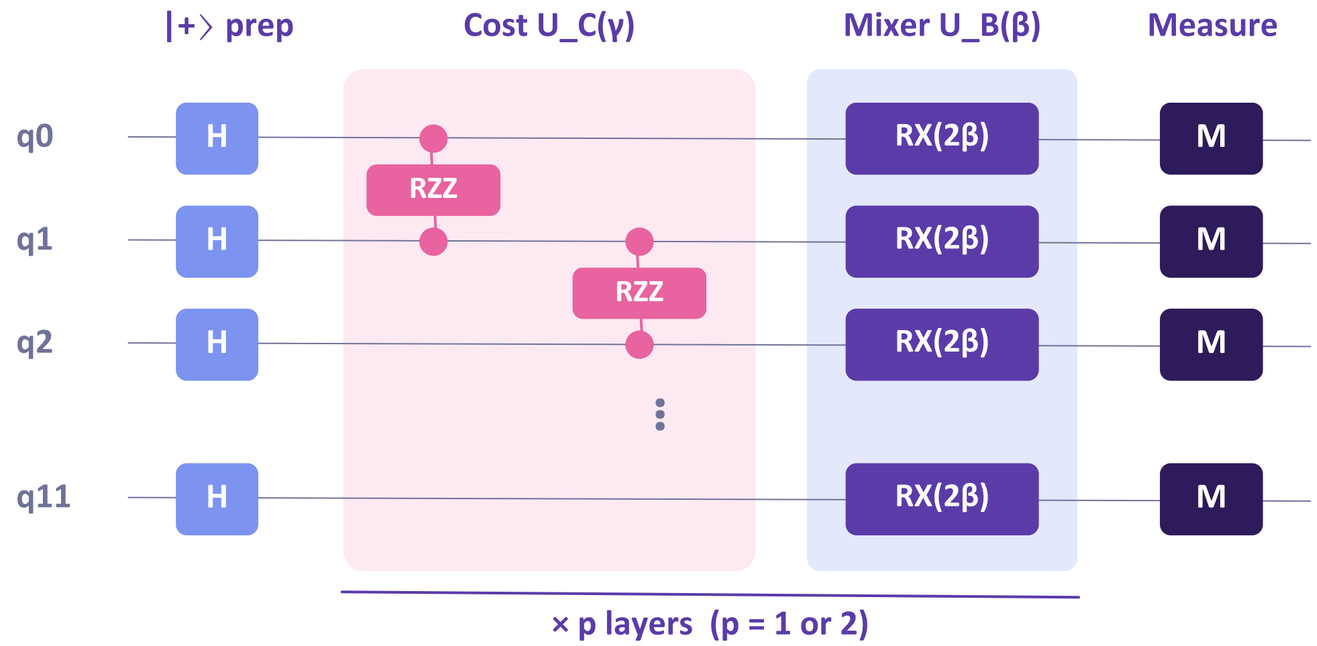
1. **Classical Baseline**
   Solves the full campus schedule (4,200 students, 180 courses) with **0 exam clashes**.
2. **Bottleneck Isolation**
   Extracts the single toughest conflict cluster (10–12 courses causing back-to-back exam fatigue).
3. **Quantum Optimization (QAOA)**
   Explores slot combinations in superposition to reduce student stress.
4. **Classical Safety Gate**
   Verifies the quantum result; accepts it **only** if it improves the schedule with 0 clashes, else falls back to baseline.

The Quantum Advantage (Why It Matters)

5,000+ → 12 binary variables → qubits

- **Massive qubit compression**
  Shrinks a 5,000+ variable problem into just 12 qubits (1 qubit = 1 course move), fitting today's IBM hardware.
- **Untangles soft conflicts**
  Targets dense knots of back-to-back exams where classical solvers often stall.
- **100% error-proof output**
  Hardware noise can't reach students: the gate guarantees a mathematically clash-free schedule.

Algorithm & Circuit Design



COBYLA / SPSA updates (γ, β) to minimise measured energy

QUANTA Algorithm

- **Classical Baseline (CP-SAT):**Solves full campus schedule with 0 clashes.
- **Isolate Bottleneck (NetworkX):**Extracts worst conflict cluster (< 12 courses).
- **1-Qubit Mapping (QUBO):** Assigns 1 qubit per course (`0` = Stay in Slot A, `1` = Move to Slot B).
- **Quantum Search (QAOA):**Explores slot combinations in superposition to minimize student fatigue.
- **Classical Repair:** Matches rooms (Hopcroft-Karp) and rolls back invalid moves.
- **Safety Gate:** Deploys only if Clashes = 0& fatigue improves; else keeps baseline.

Implementation & Tech Stack

1 Data Ingestion & Baseline

Parses 4 CSVs, solves the 4,200-student schedule with **0 hard violations** (CP-SAT).

2 Bottleneck Extraction

NetworkX conflict graph isolates the ≤ 12-course friction cluster and a candidate **Slot B**.

3 Quantum Execution

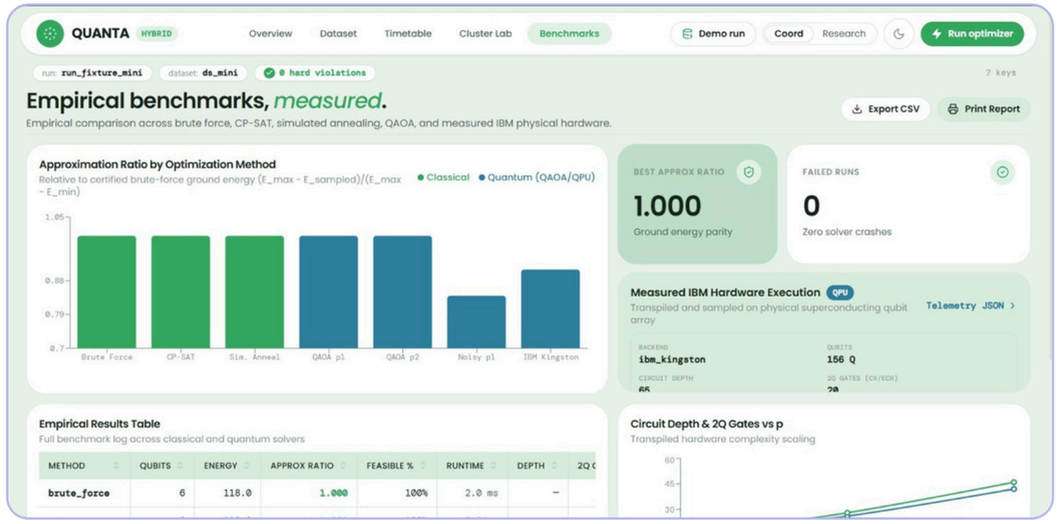
12-qubit Ising HC + **QAOA** (COBYLA / SPSA), 512–4,096 shots → **top 32** bitstrings.

4 Repair & Safety Gate

Hopcroft-Karp + rollback fix rooms; **Q-OPT** deploys only if hard = 0 and soft drops.

TECH STACK · LAYER	TECHNOLOGIES & TOOLS	CORE RESPONSIBILITY
Quantum Engine backend/quantum/	**Qiskit** · **Aer** · **Runtime** (SamplerV2, Estimator) · **IBM Kingston QPU**, 156-qubit Heron r2	QUBO→Ising, QAOA transpile (level 3, dynamical decoupling), T_1/T_2 noise sim, live QPU
Classical Engine backend/core/	**Python** · **Google OR-Tools (CP-SAT)** · **NetworkX** · **Hopcroft-Karp**	Global solve, conflict-graph clustering, room matching, zero-trust constraint audit
Backend & API backend/api/	**FastAPI** · **Pydantic v2** · **SQLite** · **Uvicorn** · **Pytest** (61 tests)	REST endpoints, schema validation, async job queue for QPU runs, telemetry caching
Frontend Console frontend/	**React** · **TypeScript** · **Vite** · **D3.js** · **Recharts** · **Bento Grid UI**	Master timetable, before/after conflict graph, live energy traces, hardware telemetry

QUANTA: Results & Benchmark Analysis



1 Key Performance Metrics

Campus demo dataset: 4,200 students, 180 courses

1.8 s
CP-SAT global baseline
Full campusschedulesolved with 0 hard violations.

412 → 380
Quantum bottleneck gain
Softpenalty cut; back-to-backfatigue resolved for 18 students.

100%
Reliability guarantee
0hardclashesacross10 seeds & 61 tests via the Acceptance Gate.

2 Multi-Solver Benchmark Comparison

6 to 12 qubits

SOLVER METHOD	BACKEND & RUNTIME	KEY ANALYSIS & BEHAVIOR
Classical Exact cpsat, brute_force	Local CPU · < 40 ms	Finds the exact optimal ground state almost instantly for small clusters (N ≤ 12).
Classical Heuristic sim_anneal	Local CPU · fast (milliseconds)	Explores local neighborhoods thermally; benchmarked against exact and quantum approximation ratios.
Quantum Ideal qaoa_ideal	Statevector simulator · several seconds	Smooth variational energy convergence (H_C expectation) closely matching the brute-force ground truth.
Quantum Noisy qaoa_noisy	Qiskit Aer (T_1/T_2 noise) · several seconds	Resilient under 10^{-2} 2-qubit gate errors and 2% readout bit-flips when paired with SPSA & classical repair.
Quantum Hardware qaoa_ibm	IBM Kingston QPU (156-qubit Heron r2) · queue time + QPU run	Validated via Level-3 transpilation & dynamical decoupling (512–4,096 shots); noisy bitstrings safely filtered by the repair gate.

3 Scientific Takeaways & Hardware Scaling

- **Circuit scaling (N = 6 → 12 qubits):** tracks transpiled circuit depth and physical 2-qubit ECR / CNOT gate counts across p = 1 and p = 2 QAOA layers.
- **Honest NISQ evaluation:** at N ≤ 12 classical solvers are faster (< 40 ms vs. seconds for QAOA). QUANTA's true breakthrough is architectural: compressing a 5,000+ variable problem into 12 qubits and safely deploying real quantum hardware with zero risk of schedule failure.

Implementation & Tech Stack

1 Data Ingestion & Baseline

Parses 4 CSVs, solves the 4,200-student schedule with **0 hard violations** (CP-SAT).

2 Bottleneck Extraction

NetworkX conflict graph isolates the ≤ 12-course friction cluster and a candidate **Slot B**.

3 Quantum Execution

12-qubit Ising HC + **QAOA** (COBYLA / SPSA), 512–4,096 shots → **top 32** bitstrings.

4 Repair & Safety Gate

Hopcroft-Karp + rollback fix rooms; **Q-OPT** deploys only if hard = 0 and soft drops.

TECH STACK · LAYER	TECHNOLOGIES & TOOLS	CORE RESPONSIBILITY
Quantum Engine backend/quantum/	**Qiskit** · **Aer** · **Runtime** (SamplerV2, Estimator) · **IBM Kingston QPU**, 156-qubit Heron r2	QUBO→Ising, QAOA transpile (level 3, dynamical decoupling), T_1/T_2 noise sim, live QPU
Classical Engine backend/core/	**Python** · **Google OR-Tools (CP-SAT)** · **NetworkX** · **Hopcroft-Karp**	Global solve, conflict-graph clustering, room matching, zero-trust constraint audit
Backend & API backend/api/	**FastAPI** · **Pydantic v2** · **SQLite** · **Uvicorn** · **Pytest** (61 tests)	REST endpoints, schema validation, async job queue for QPU runs, telemetry caching
Frontend Console frontend/	**React** · **TypeScript** · **Vite** · **D3.js** · **Recharts** · **Bento Grid UI**	Master timetable, before/after conflict graph, live energy traces, hardware telemetry

QUANTA: Challenges & Future Scope

Current Challenges & NISQ Limitatior

Qubit capacity bottleneck
Full campus needs 5,000+ binary variables—far beyond today's 100–156-qubit processors.

Hardware noise & decoherence
T_1 = 120 µs, T_2 = 90 µs, 2-qubit CNOT errors ≈ 10^{-2}, and 2% readout bit-flips.

Runtime latency vs. classical
For N ≤ 12, CP-SAT and brute force solve in <40 ms; QAOA takes seconds plus QPU queue time.

Single-hop move restriction
Each course can currently stay in baseline Slot A or move to one alternate Candidate Slot B.

Future Scope & Roadmap

Scale cluster sizes
Expand beyond ≤12 courses toward 50–100+ qubits as hardware fidelity and gate error rates improve.

Multi-slot quantum encoding
Move beyond Slot A/B binary perturbations to qudit or multi-qubit encodings for 4+ candidate slots.

Error mitigation & deeper QAOA
Integrate ZNE and PEC through Qiskit Runtime to support deeper circuits (p > 2) despite noise.

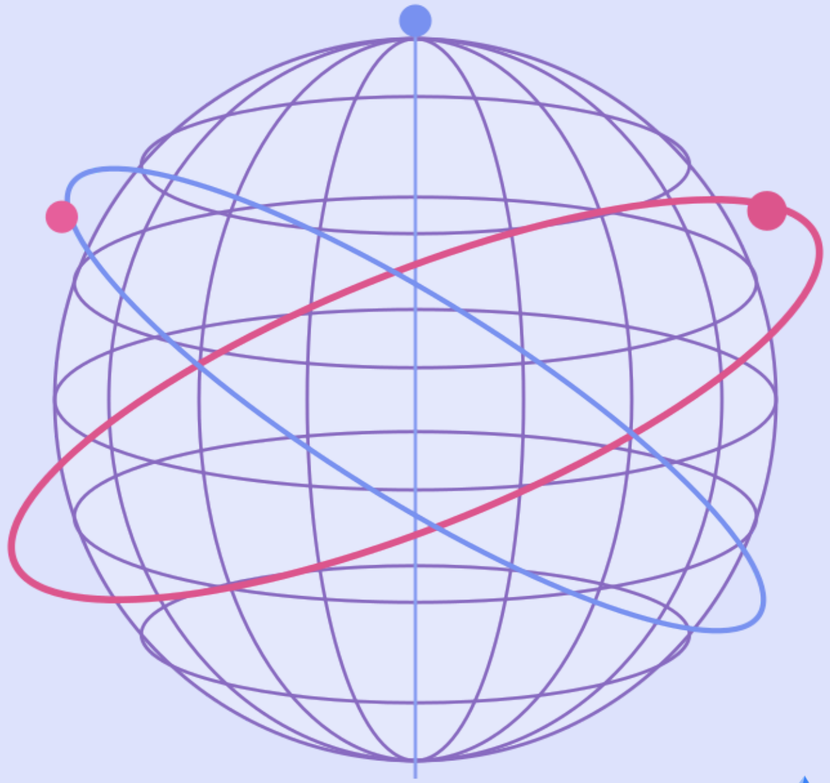
Parallel clusters & enterprise APIs
Optimize disjoint clusters concurrently; generalize the engine to hospital shifts, airline crews, and cloud GPU allocation.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY
Quantify Click here

DEMO VIDEO LINK
Youtube video Click here